THE IDEAL OF THE TARGET ACTING ON A PUSHFORWARD ALONG AN AFFINE MORPHISM, WITH THE NOTATION SPELLED OUT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The scheme case of the projection formula for an ideal along an affine morphism. Cohomology of Schemes, label coherent-lemma-affine-morphism-projection-ideal (invoked at proof line 3409). After the reduction to an affine base it is exactly the asserted equality $\mathcal{I}f_*\mathcal{F} = f_*(f^{-1}\mathcal{I}\mathcal{F})$.
- Exactness of the inverse image functor, which makes the inverse image of an ideal a sheaf of ideals. Standard, used at proof lines 3396-3401. It shows that $f^{-1}\mathcal{I}$ is a sheaf of ideals of $f^{-1}\mathcal{O}_X$, acting on $\mathcal{F}$ through $f^\sharp$, so that $f^{-1}\mathcal{I}\mathcal{F}$ makes sense.

Source reference. Original lemma 01YP: sources/stacks-chow/tags.txt, source lines 2564–2564. Navigation pointer for \ref{coherent-lemma-affine-morphism-projection-ideal} used inside the counted span (line 3413); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent sheaf on Y. Let $\mathcal{I}$ be a quasi-coherent sheaf of ideals on X. If f is affine then $\mathcal{I}f_*\mathcal{F} = f_*(f^{-1}\mathcal{I}\mathcal{F})$ (with notation as explained in the proof).*

Proof. The notation means the following. Since f^{-1} is an exact functor we see that $f^{-1}\mathcal{I}$ is a sheaf of ideals of $f^{-1}\mathcal{O}_X$. Via the map $f^\sharp : f^{-1}\mathcal{O}_X \to \mathcal{O}_Y$ on $Y_{étale}$ this acts on $\mathcal{F}$. Then $f^{-1}\mathcal{I}\mathcal{F}$ is the subsheaf generated by sums of local sections of the

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form as where a is a local section of $f^{-1}\mathcal{I}$ and s is a local section of $\mathcal{F}$. It is a quasi-coherent $\mathcal{O}_Y$-submodule of $\mathcal{F}$ because it is also the image of a natural map $f^*\mathcal{I} \otimes_{\mathcal{O}_Y} \mathcal{F} \to \mathcal{F}$.

Having said this the proof is straightforward. Namely, the question is étale local on X and hence we may assume X is an affine scheme. In this case the result is a consequence of the corresponding result for schemes, see Cohomology of Schemes, Lemma 01YP. □